

Calibration of a Lifecycle Model of Housing and Fertility

External inputs, internal moments, and model fit

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Calibration objective

The model must jointly reproduce four lifecycle choices:

- completed family size;
- tenure and occupied housing;
- liquid saving and entry wealth;
- late-life estates and housing retention.

The central mechanism is that children raise housing needs, while access to owner housing depends on wealth and the down payment.

Calibration therefore uses variation across age, tenure, family status, housing size, and wealth.

Empirical discipline

CPS

- Completed fertility
- Completed childlessness

ACS

- Ownership by age and family status
- Rooms by tenure and family size
- Aggregate occupied rooms

PSID

- Entrant wealth
- Young liquid wealth
- Late-life estates and nonhousing wealth

Each dataset disciplines a different margin, but all parameters are estimated jointly.

Source: Circulated calibration note, Sections 1.1–1.3.

Lifecycle timing

A model period is four years.

18	fertile years	66	85
entry	family-size choice	retirement	maximum age

- Survival is certain before retirement and follows the 2023 Social Security life table thereafter.
- Dependent children remain at home for 18 years on average: maturity probability $2/9$ per period.
- Risk aversion is fixed at $\sigma = 2$.

Preferences and family needs

Children affect both the utility from family size and the resources required by the household.

Consumption and housing requirements are

$$\bar{c}(n_s) = \bar{c}_0 + \bar{c}_n n_s, \quad \bar{h}(n_s) = \bar{h}_0 + \bar{h}_{jump} \mathbf{1}\{n_s \geq 1\} + \bar{h}_n n_s.$$

- ψ_{child} governs the direct utility value of children.
- κ_F governs dispersion in completed-family-size choices.
- $\bar{h}_{jump}$ isolates the first-child increase in space needs.

Source: Circulated calibration note, Tables 2–3 and “Consumption needs and family size.”

Housing products

Renting

$$h^R \in [0, 6].$$

Renters choose housing continuously up to six rooms, the 90th percentile of the matched ACS rental distribution.

The tenure-specific menus make family-sized housing lumpy and link fertility to tenure choice.

Owning

$$H \in \{2, 4, 6, 8, 10\}.$$

Owners choose from five discrete house sizes that summarize the observed room distribution.

Source: Circulated calibration note, "Housing products and supply."

Housing finance

Mortgage debt cannot exceed $\phi = 0.80$ of house value, so purchase requires

$$(1 - \phi)PH = 0.20PH$$

in equity.

- Real liquid return: 2.0%
- Housing depreciation: 1.1%
- Property tax: 1.0%
- Selling cost: 6%
- Labor-income tax: 17.9%
- No separate payment-to-income limit

Source: Circulated calibration note, "Housing finance and transaction costs."

Income process

Earnings combine a deterministic age profile with persistent idiosyncratic risk:

$$\log z_{t+1} = \rho_z \log z_t + \varepsilon_{t+1}, \quad \rho_z = 0.960, \quad \sigma_z = 0.0645.$$

The annual process is sampled at the four-year model frequency and approximated by a five-state Rouwenhorst chain.

The persistence is consistent with the upper end of PSID estimates. The innovation variance is deliberately below the empirical range and the quantitative results are conditional on it.

Source: Circulated calibration note, "Income process."

Housing supply

Aggregate housing services are supplied elastically, with elasticity

$$\eta = 1.75,$$

based on the population-weighted metropolitan estimate in Saiz (2010).

The scale of housing supply, $\bar{H}$, is estimated internally so that the model matches aggregate occupied rooms.

The calibration therefore separates:

- an externally assigned supply elasticity; and
- an internally estimated supply level.

Entry wealth and bequests

Households enter at age 18 with a draw from the PSID net-worth-to-income distribution of childless renters aged 18–24.

The benchmark imposes

$$\theta_n = 0,$$

so the bequest motive does not vary directly with the number of children.

Children can still affect estates indirectly through housing needs, tenure, and lifecycle saving.

Source: Circulated calibration note, “Entry wealth and bequests.”

Parameters set outside estimation

Parameter	Value	Basis
Period; entry; retirement; max age	4; 18; 66; 85	Model timing
Child duration; risk aversion	18 years; $\sigma = 2$	Maintained restrictions
Return; depreciation; property tax	2.0%; 1.1%; 1.0%	External evidence / maintained tax
Financed share; selling cost	0.80; 6%	Housing-finance evidence
Income persistence; innovation s.d.	0.960; 0.0645	PSID literature / provisional
Owner menu; renter maximum	$\{2, 4, 6, 8, 10\}; 6$	Matched ACS distributions
Supply elasticity	1.75	Saiz (2010)
Entrant wealth; θ_n	PSID distribution; 0	Data / maintained restriction

Source: Circulated calibration note, Table 1.

Internal estimation

Fourteen parameters are estimated jointly using fifteen moments.

Saving

β_{annual}

Housing

$\alpha_c, \bar{h}_0,$
 $\bar{h}_{jump}, \bar{h}_n$

Family size

$\bar{c}_0, \bar{c}_n,$
 ψ_{child}, κ_F

Tenure / estates

$\chi_O, \bar{H},$
 $\theta_0, \theta_1, \kappa_T$

The blocks organize the economics; the estimator chooses all parameters simultaneously because most moments load on several margins.

Source: Circulated calibration note, Table 2 and Section 1.2.

Saving block

The annual discount factor is estimated at

$$\beta_{annual} = 0.9912.$$

It is disciplined jointly by wealth at three lifecycle stages:

- liquid wealth among young childless renters;
- the share of older households with substantial nonhousing wealth;
- the late-life median estate.

The estimate is a joint calibration outcome, not a direct measure of household patience.

Source: Circulated calibration note, "Saving."

Housing-demand block

Estimates

$$\begin{aligned}\bar{h}_0 &= 0.3915, \\ \bar{h}_{jump} &= 1.6021, \\ \bar{h}_n &= 0.1644.\end{aligned}$$

Empirical discipline

- Childless-renter mean rooms
- First-child increase in rooms
- Rooms gap between larger and smaller families

The first child creates the largest discrete increase in required space.

Source: Circulated calibration note, "Housing demand, tenure, and supply."

Tenure and supply block

Tenure demand

$$\chi_O = 1.1133, \quad \kappa_T = 0.0100.$$

Ownership rates by age and family status discipline the owner-service premium and tenure-choice dispersion.

Tenure and housing-size moments identify a common block because ownership changes both access and the menu of available space.

Housing supply

$$\bar{H} = 8.6456.$$

Aggregate occupied rooms disciplines the supply scale; the owner-renter rooms gap helps discipline housing demand.

Source: Circulated calibration note, "Housing demand, tenure, and supply."

Family-size block

Needs

$$\bar{c}_0 = 1.2596,$$
$$\bar{c}_n = 0.3929.$$

Young childless wealth is particularly informative about the baseline requirement; family-composition differences discipline the incremental cost.

Costs and tastes are estimated jointly because the same fertility outcomes respond to both.

Preferences

$$\psi_{child} = 0.1979,$$
$$\kappa_F = 2.0434.$$

Completed fertility and childlessness discipline the utility value of children and dispersion in family-size choices.

Source: Circulated calibration note, "Consumption needs and family size."

Bequest block

The strength and wealth dependence of the bequest motive are

$$\theta_0 = 0.3118, \quad \theta_1 = 0.3973.$$

- The median late-life estate is particularly informative about θ_0 .
- Older nonhousing wealth and old-age ownership add discipline for θ_1 .
- The same moments also respond to saving and tenure, so the bequest block is estimated jointly with them.

Source: Circulated calibration note, "Bequests."

Family size and family housing

Moment	Data	Model	Gap
Completed fertility	1.918	2.034	+0.116
Childlessness rate	0.188	0.189	+0.001
Family ownership gap	0.168	0.219	+0.052
First-child housing increment, rooms	0.664	0.681	+0.016
Rooms gap, 3+ versus 1–2 children	0.368	0.607	+0.240

The model closely matches childlessness and the first-child housing response. It overstates the separation in ownership and rooms between larger and smaller families.

Source: Circulated calibration note, Table 3.

Tenure, housing, and wealth

Moment	Data	Model	Moment	Data	Model
Ownership, ages 30–55	.575	.658	Ownership, ages 25–34	.341	.274
Childless-renter rooms	3.805	3.963	Ownership, ages 65–75	.764	.954
Childless-owner share, 6+ rooms	.596	.760	Mean occupied rooms	5.780	5.673
Owner-renter rooms gap	2.419	2.323	Young liquid wealth / income	.179	.328
Median estate / income	6.501	6.431	Older nonhousing wealth share	.608	.610

Housing quantities and late-life wealth levels are close. The largest misses are excessive old-age ownership, too many childless owners in large homes, and excess liquid wealth among young renters.

Source: Circulated calibration note, Table 3 and discussion of fit.

Summary

- The calibration combines externally assigned timing, finance, income, and supply parameters with 14 internally estimated parameters.
- Fifteen moments jointly discipline fertility, housing needs, tenure, saving, and bequests.
- The model matches childlessness, the first-child housing response, aggregate rooms, and targeted late-life wealth levels.
- The main tensions are ownership by age, large-home selection, young liquid wealth, and the thin upper tail of estates.

The current calibration is designed for quantitative exercises on family housing needs, tenure, and the down-payment constraint; conclusions that depend on late-life portfolio composition remain outside its scope.